

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
DLA 13

Histology of the Respiratory System

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Assigned Reading

Histology – A Text and Atlas

Pawlina 9th Edition

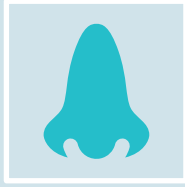
Chapter 19: Respiratory System Pg. 662-697

<https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=257429520&bookid=3290>

OBJECTIVES

SOM.MK.1.BPM1.1.CPR.3.HCB.0561	List the divisions of the respiratory system.
SOM.MK.1.BPM1.1.CPR.3.HCB.0562	Describe the components of each division of the respiratory system.
SOM.MK.1.BPM1.1.CPR.3.HCB.0563	Describe the 5 cell types of the respiratory epithelium.
SOM.MK.1.BPM1.1.CPR.3.HCB.0564	Describe the histological structure of the vestibular and ventricular folds of the larynx
SOM.MK.1.BPM1.1.CPR.3.HCB.0565	State the function(s) of the larynx
SOM.MK.1.BPM1.1.CPR.3.HCB.0566	Predict the outcome of defective cilia in Kartagener's Syndrome in the respiratory system.
SOM.MK.1.BPM1.1.CPR.3.HCB.0567	Describe the histopathogenesis of cystic fibrosis specific to the respiratory system.

OVERVIEW OF THE RESPIRATORY SYSTEM



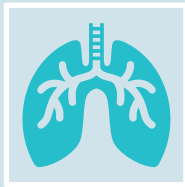
Air Conduction

Transport air
Condition air (warm, moisten)
Olfaction, Phonation



Air Filtration

Air clearance
Immune function

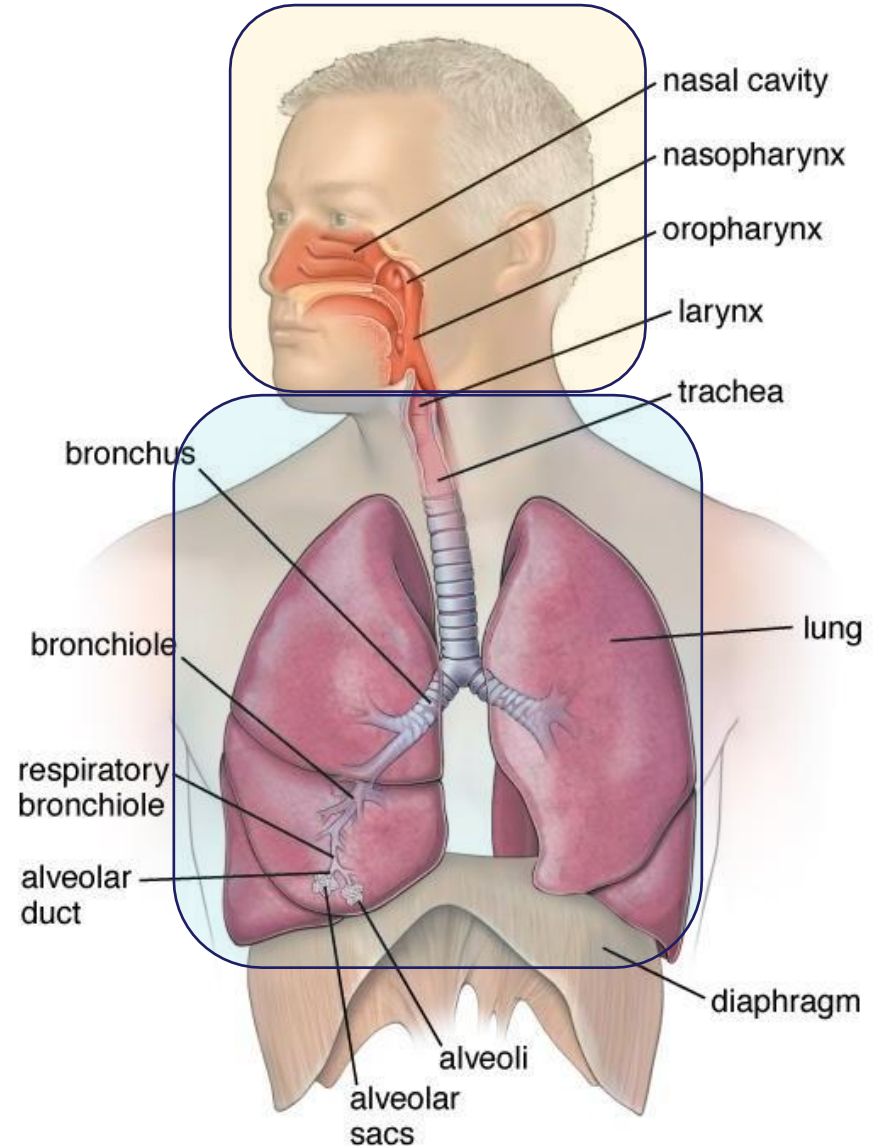


Respiration (Gaseous exchange)

Blood-gas barrier

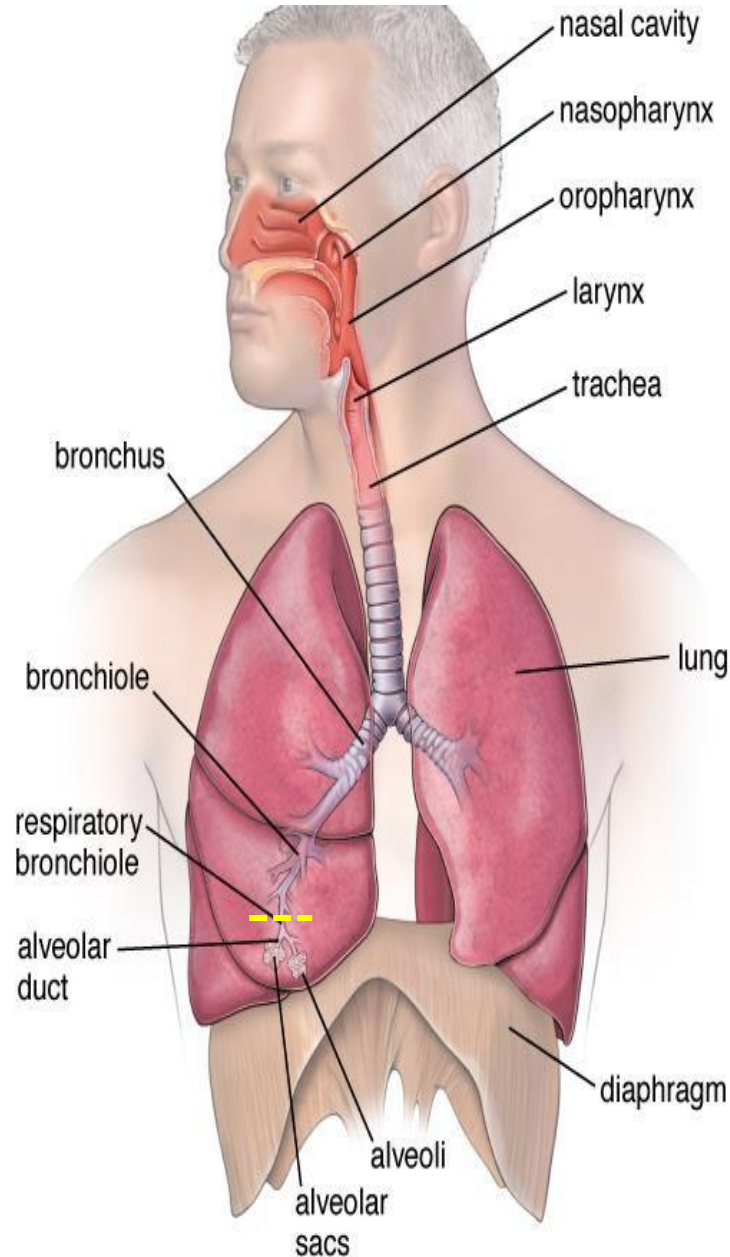
OVERVIEW OF THE RESPIRATORY SYSTEM

- **Upper Part**
 - Nasal cavity - pharynx
- **Lower Part**
 - Larynx and inferior organs
- **Conducting Portion**
 - Extrapulmonary
 - Nasal cavity to primary bronchi
 - Intrapulmonary
 - Secondary bronchi to terminal bronchioles
- **Respiratory Portion**
 - Respiratory bronchioles onward

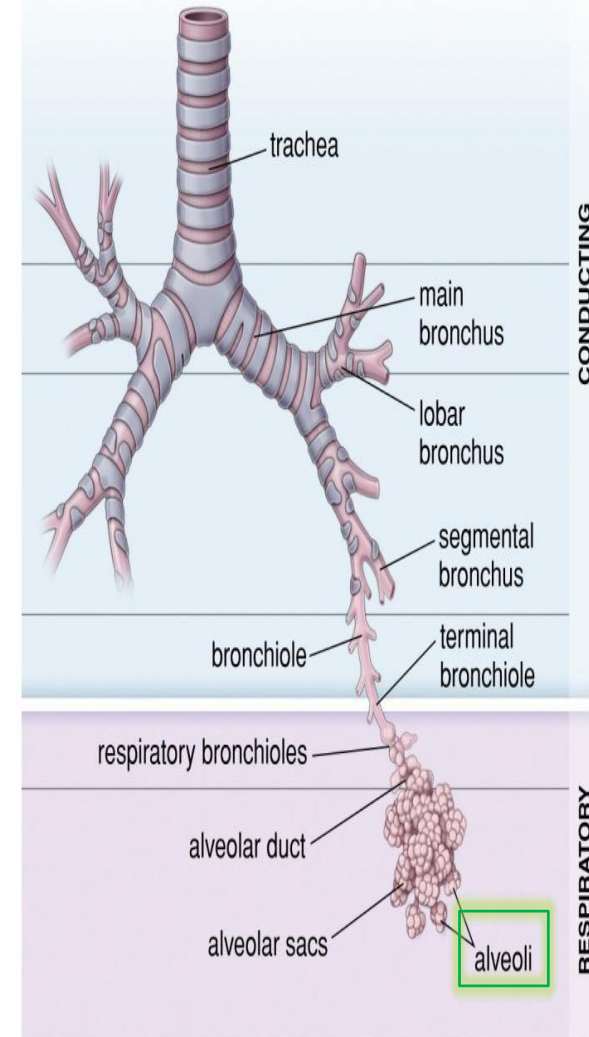


RESPIRATORY SYSTEM COMPONENTS & FUNCTIONS

1. *Olfaction*
 - Olfactory mucosa in nasal cavity
2. *Phonation*
 - Vocal folds
3. *Conduction and conditioning of air*
 - Transport of air to respiratory portion
 - Warmth
 - Air clearance
4. *Gaseous exchange:*
 - **Respiratory portion** → blood-gas barrier
5. *Endocrine* → hormones
 - Neuroendocrine cells
6. *Immune functions*
 - Bronchial associated lymphoid tissue
 - Alveolar macrophages

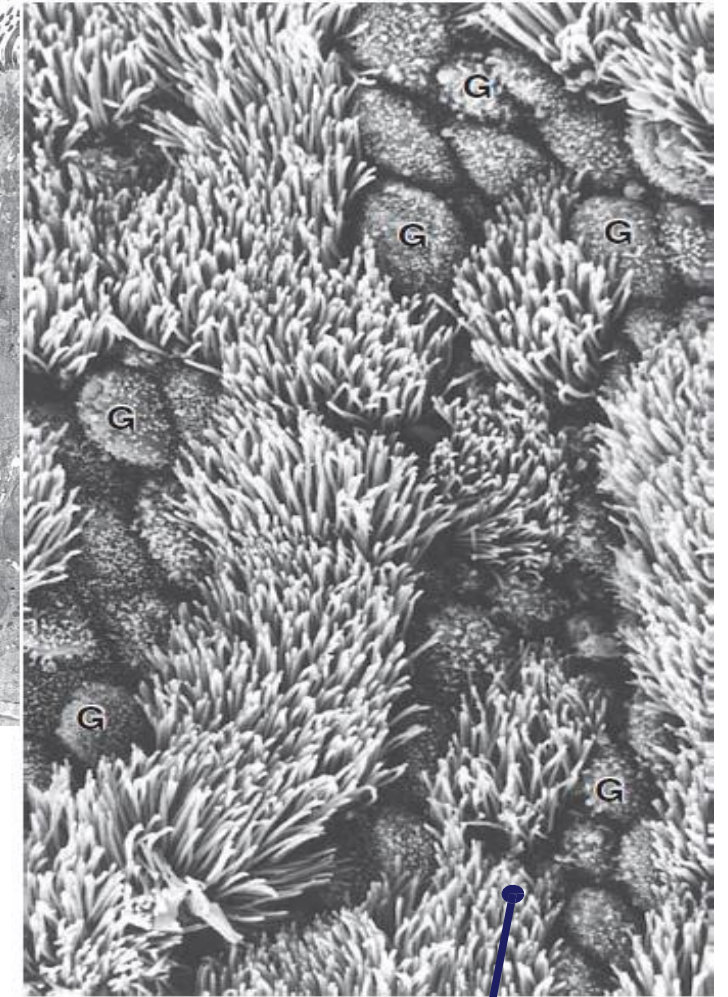
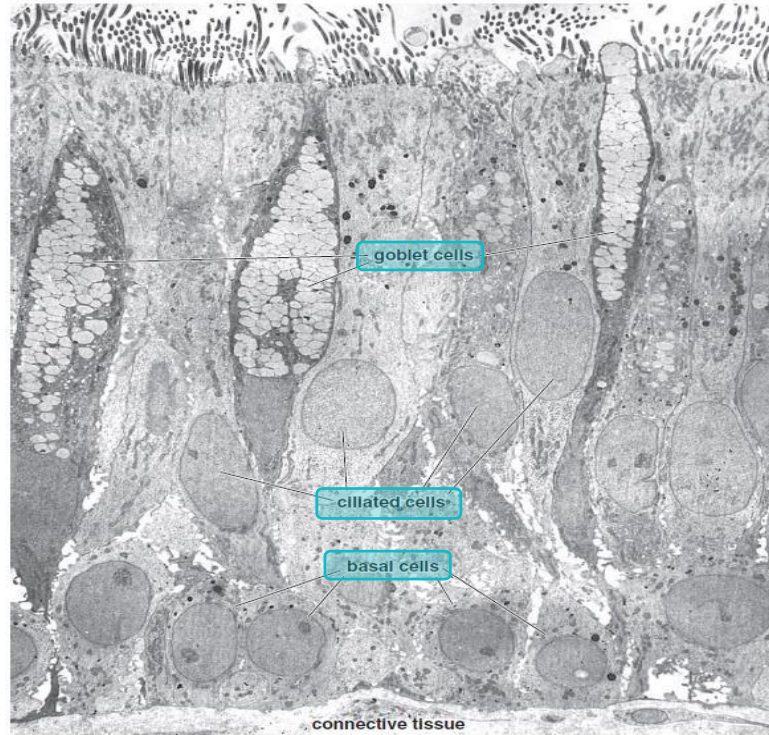
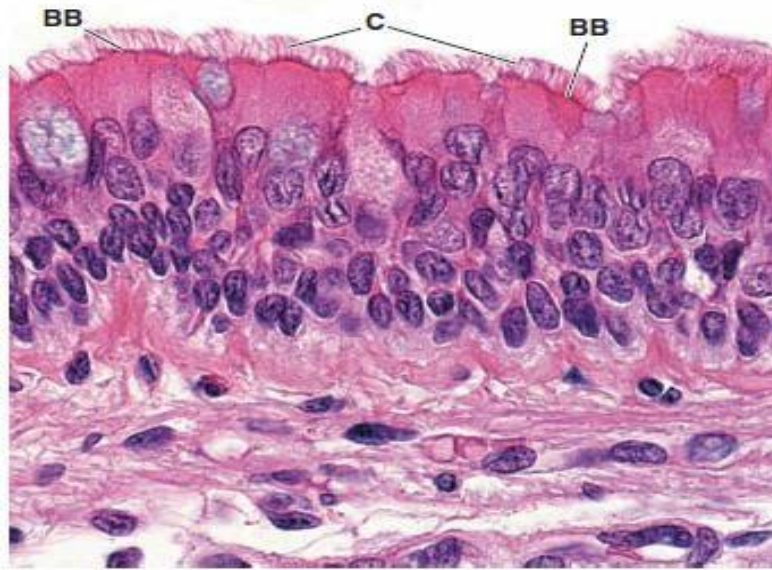


Infraglottic Airways



RESPIRATORY EPITHELIUM

Cell type	Histological features	Function	Clinical Notes
Ciliated columnar cells (most numerous)	- Columnar cells, extend to apical surface - Each cell has ~250 cilia	Sweeping motion helps expel particles trapped in mucus	Primary ciliary dyskinesia in Kartegener's syndrome
Mucus (Goblet) cells	- Interspersed among ciliated cells - Extend full thickness of the epithelium - Clear cytoplasm, no cilia	Synthesize+ secrete mucin granules → forms protective barrier	Increased in smokers and chronic inflammation
Brush cells	- Columnar cells, extend to apical surface - Have blunt microvilli	Basal surface synapse with afferent nerves → sensory function	Primary cells affected in small cell carcinoma of the lung
Small granule or Kulchitsky cells	- Sparsely dispersed among other cells - Similar to, but larger, than basal cells - Contains secretory granules	- Enteroendocrine cells - Secrete catecholamines, ADH, ACTH, serotonin, calcitonin and bombesin	
Basal cells	- Lie on basement membrane - Nuclei form row close to basal lamina - Makes no contact with apical surface	Stem cells that regenerate all other cell types	



G – Goblet cells/ Ciliated cells
Surface view

- Pseudostratified ciliated columnar epithelium with goblet cells
- Lines most of the conducting part of the respiratory system

C – Ciliated columnar cells

BB – Basal body

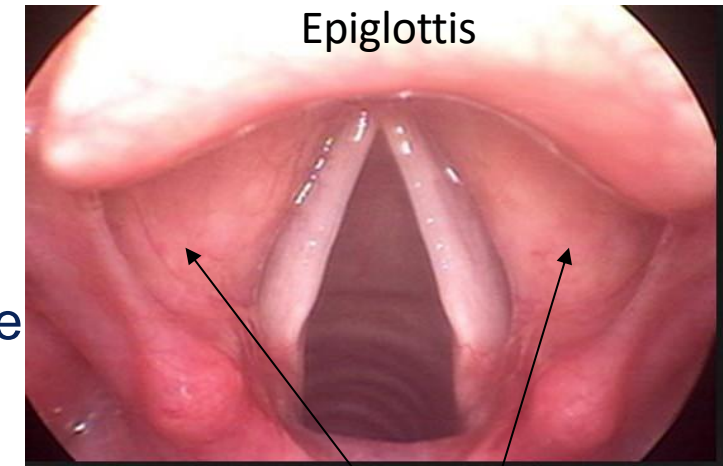
LARYNX

Structure and Function of the Larynx

- Passageway for air between oropharynx and trachea.
- A complex tubular region of the respiratory system.
- Formed by irregularly shaped plates of hyaline and elastic cartilage
 - Epiglottis
 - Vocal processes of arytenoid cartilages

Vocal Folds (True Vocal Cords)

- The vocal folds are two mucosal folds projecting into the laryngeal lumen.
- Functions:
 - Control airflow through the larynx.
 - Vibrate to produce sound.
- Oriented anteroposteriorly, forming the lateral boundaries of the rima glottidis.
- Each vocal fold contains:
 - A supporting ligament
 - The vocalis muscle (skeletal muscle)
- Intrinsic laryngeal muscles:
 - Join adjacent cartilaginous plates.
 - Responsible for:
 - Creating tension in vocal folds.
 - Opening and closing the glottis.



Vestibular fold

Extrinsic Laryngeal Muscles

- Insert on laryngeal cartilages but originate from **outside the larynx**.
- Function: **Move the larynx during swallowing (deglutition).**

LARYNX cont..

Ventricular Folds (False Vocal Cords)

- Located above the true vocal folds.
- Form the ventricular folds, also known as false vocal cords.
- They:
 - Lack intrinsic muscle.
 - Do not participate in phonation.
- **The ventricle:**
 - An elongated recess located above the vocal folds.
 - Lies directly below the ventricular folds.
- Ventricular folds + ventricle:
 - Important for sound resonance.

LARYNX cont ...

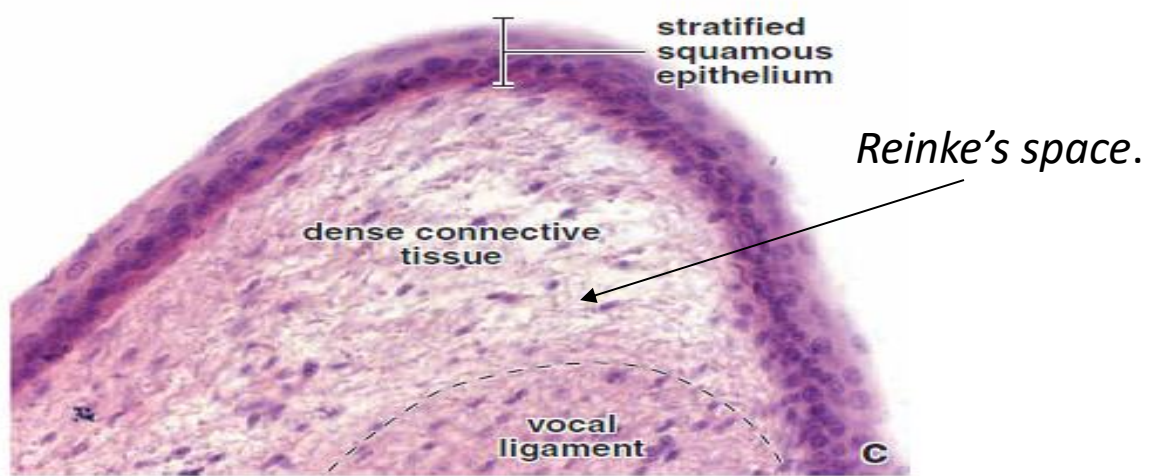
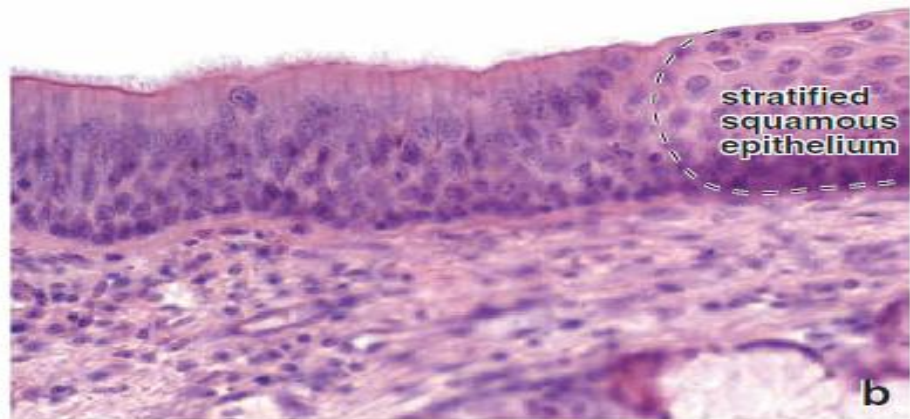
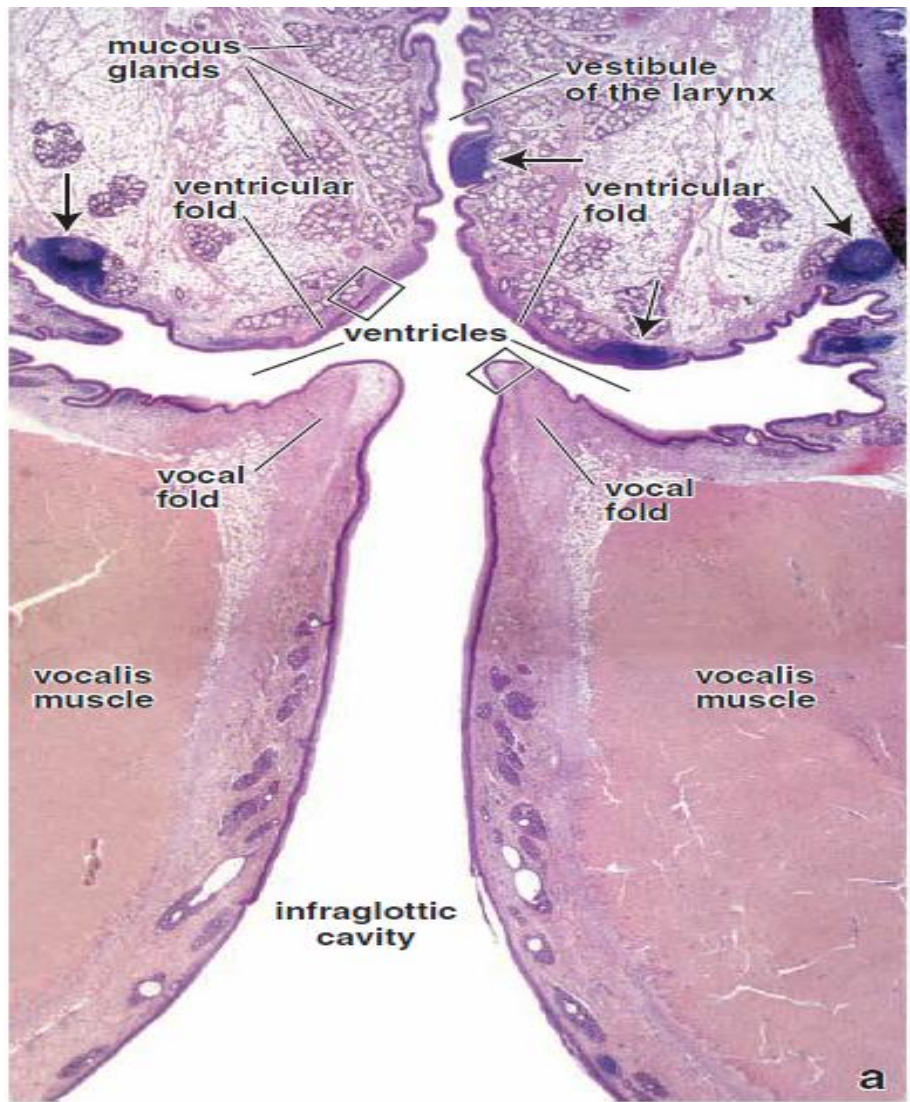
Epithelial Lining of the Larynx

- Larynx contains **two types of epithelium**:
 - **Stratified squamous epithelium**
 - **Ciliated pseudostratified columnar epithelium**
- **Specific locations**:
- **Vocal cords**: stratified squamous epithelium
 - Protects mucosa from abrasion caused by rapidly moving air.
- **Most of the epiglottis**: also stratified squamous epithelium.
- **Rest of the larynx**: lined by **ciliated pseudostratified columnar epithelium** (respiratory epithelium).

Laryngeal Connective Tissue

- Contains **mixed mucoserous glands**.
- These glands secrete through **ducts onto the laryngeal surface**.

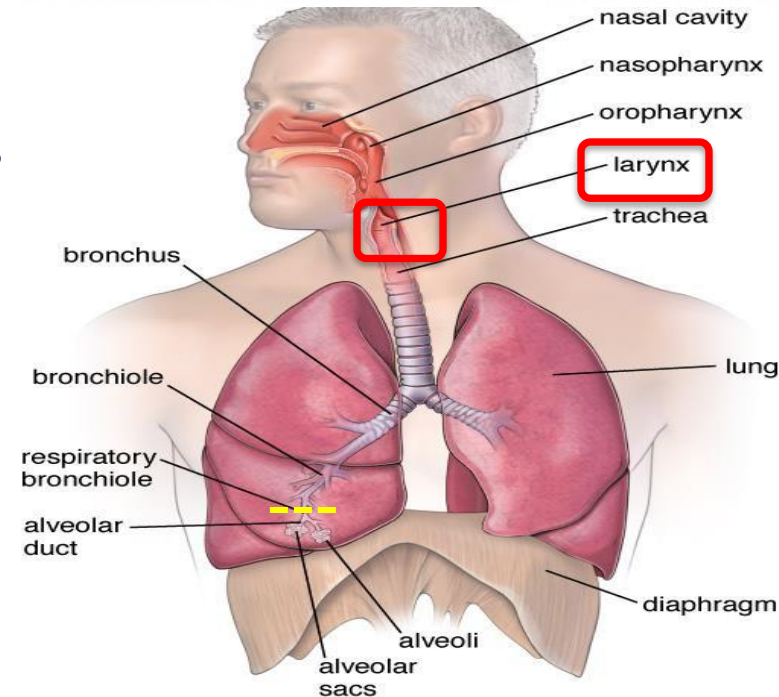
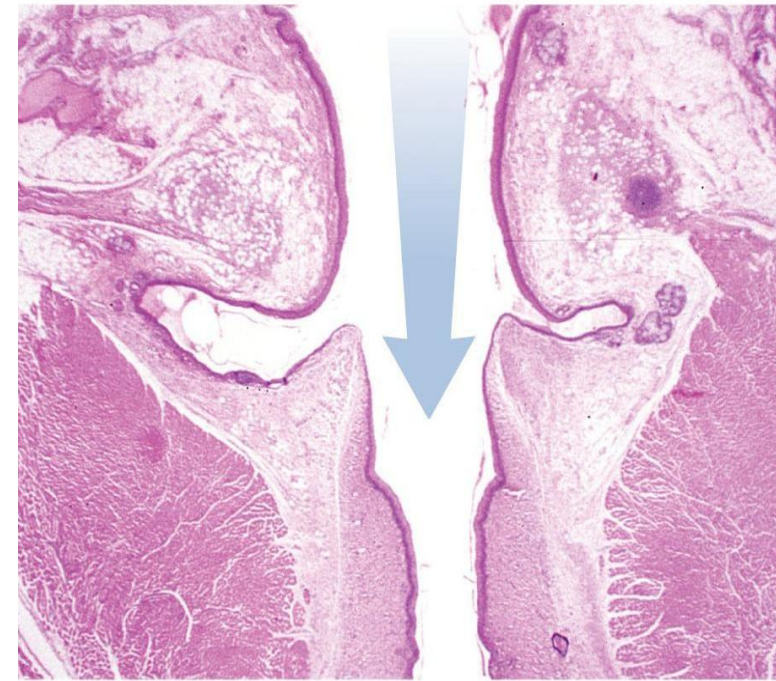
Photomicrograph of the larynx



LARYNX

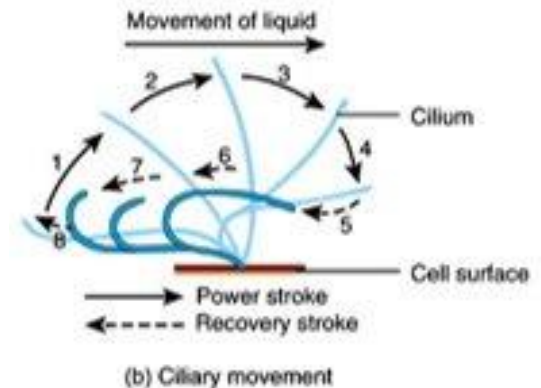
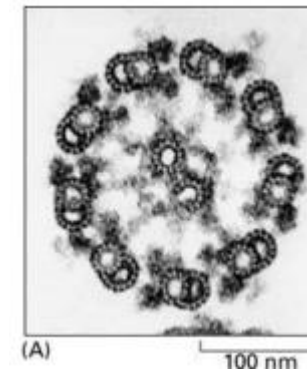
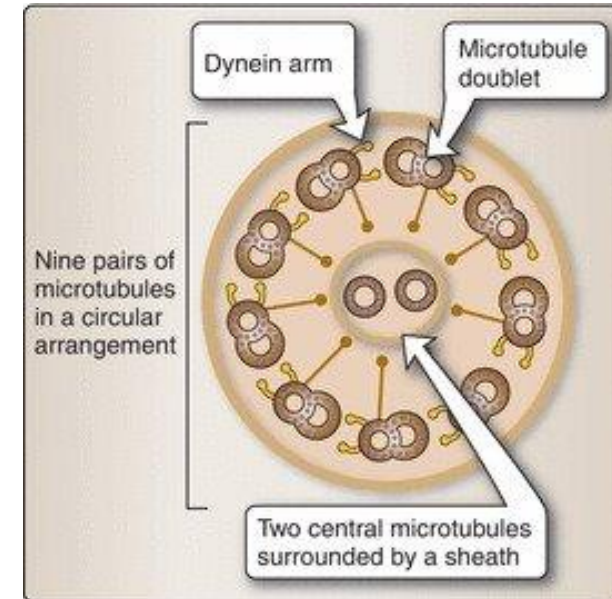
Functions of the Larynx:

1. Breathing: acts as a passageway for air to move from the throat (pharynx) to the trachea and lungs.
2. Sound Production: vocal cords vibrate to produce sound when air passes through them (essential for speaking, singing, and other vocalizations).
3. Protection of the Airway: epiglottis acts as a flap that covers the trachea during swallowing to prevent food and liquids from entering the airway.
4. Cough Reflex: helps trigger the cough reflex to expel any foreign particles or irritants that enter the airway.
5. Regulation of Airflow: controls the flow of air into the lungs, which is crucial for effective breathing and speech modulation.



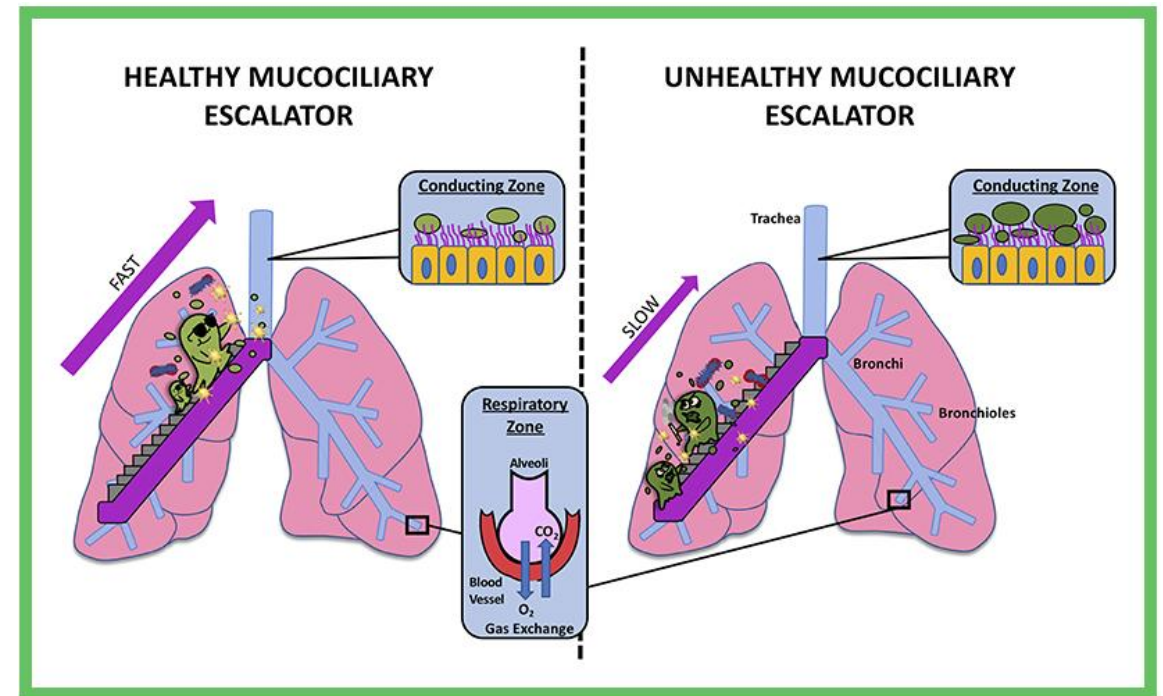
KARTAGENER'S SYNDROME

- A hereditary disorder (AR) characterized by defective ciliary motility due to structural abnormalities (commonly defective dynein arms).
- This impairs mucociliary clearance and affects organ laterality.
- Symptoms: frequent lung and sinus infections, scarring of lung tissue, sterility in males (also situs inversus)
- Defect: mutation in genes that code for ciliary proteins (dynein)= abnormal ciliary movement



KARTAGENER'S SYNDROME

‘Mucociliary escalator’: Cilia in the epithelial cells help to sweep the mucus collection in the tract towards the upper respiratory tract



Failure in the sweeping motion of cilia can result in stagnation of secretion (bronchiectasis), decreasing surface availability for gas exchange, resulting in shortness of breath. The addition of viral or bacterial infections causing inflammation of lung tissue (pneumonia) resulting into fever.

CYSTIC FIBROSIS (MUCOVISCIDOSIS)

Causes

- CFTR (cystic fibrosis transmembrane conductance regulator) gene mutation causing defective/absent Cl ion transport = increased viscosity of secretions with obstruction of excretory ducts.
- Inheritance: autosomal recessive disorder

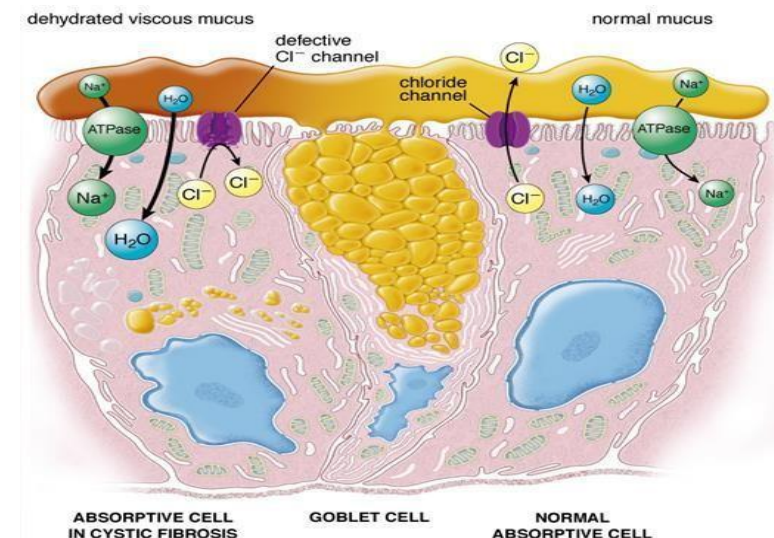
Symptoms

Respiratory: accumulation of unusually thick, viscous mucus, bronchiolar obstruction (with subsequent bronchiolar wall thickening and alveolar degeneration), frequent respiratory tract infections.

- Digestive: thick mucus can block ducts the pancreas preventing digestive enzymes from reaching the intestines. Leads to poor growth, malnutrition, and fatty stools.
- Other: CF can also affect liver, intestines, and reproductive organs. Symptoms may include sinus infections, clubbing of the fingers and toes, and infertility in males.

Diagnosis

- Sweat Test: Measures the concentration of salt in sweat. High levels indicate CF1
- Genetic Testing: Identifies mutations in the CFTR gene1



CYSTIC FIBROSIS

- Autosomal recessive genetic disorder affecting epithelial ion transport.

Cause

- Mutation in the **CFTR gene** (Cystic Fibrosis Transmembrane Conductance Regulator)
- CFTR is a **chloride channel** found in the apical membrane of epithelial cells in the lungs, pancreas, intestines, sweat glands, and reproductive tract.
 - It is unique because it works as both: **Cl⁻ ion channel** (ATP-gated channel) and **regulator of other channels and transporters**
 - **As a Chloride Channel** CFTR allows **chloride ions (Cl⁻)** to move **out** of epithelial cells **into the lumen**.
 - When CFTR transports Cl⁻ outward:
 - **Sodium (Na⁺)** follows passively (to maintain electrical balance).
 - **Water follows sodium** (osmosis).
 - CFTR inhibits the **epithelial sodium channel (ENaC)**.
 - So **less sodium moves from the lumen into the cell**
 - Water stays in the mucus → mucus stays **hydrated and thin**

- **CFTR in Sweat Glands (special case)**
 - In sweat glands, CFTR functions **differently**.
 - Sweat is produced in the coil portion.
 - As sweat moves up the duct, **CFTR reabsorbs Cl^-** , and ENaC reabsorbs Na^+ .
- **Normal CFTR → salt is reabsorbed → sweat is not salty**
- **CFTR defect → salt stays in sweat → “salty skin”**
- **Defective CFTR (Cystic Fibrosis)**
 - ↓ Cl^- secretion
 - ↑ Na^+ absorption
 - ↑ Water absorption into cells
 - → **Thick, sticky, dehydrated mucus**
 - Sweat glands cannot reabsorb NaCl → **salty sweat**

Major Organ Systems Affected

Respiratory System

- Thick mucus obstructs airways.

Clinical effects:

- Chronic cough
- Recurrent bacterial infections (S. aureus, Pseudomonas, Burkholderia)
- Bronchiectasis
- Nasal polyps
- Digital clubbing
- **2. Gastrointestinal & Pancreatic**
 - Thick secretions block pancreatic ducts → **pancreatic insufficiency.**
- **Consequences:**
 - Malabsorption of fats and proteins
 - Steatorrhea
 - Failure to thrive (children)
 - Fat-soluble vitamin deficiency (A, D, E, K)

- **3. Hepatobiliary**

- Bile duct obstruction → risk of **biliary cirrhosis.**

- **4. Reproductive System**

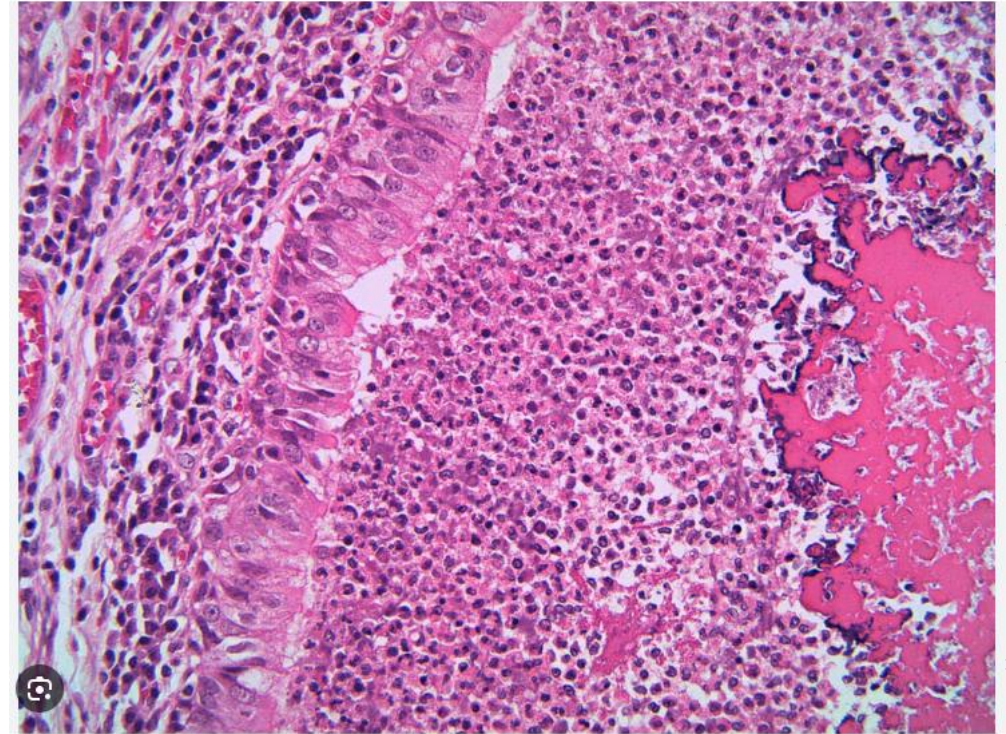
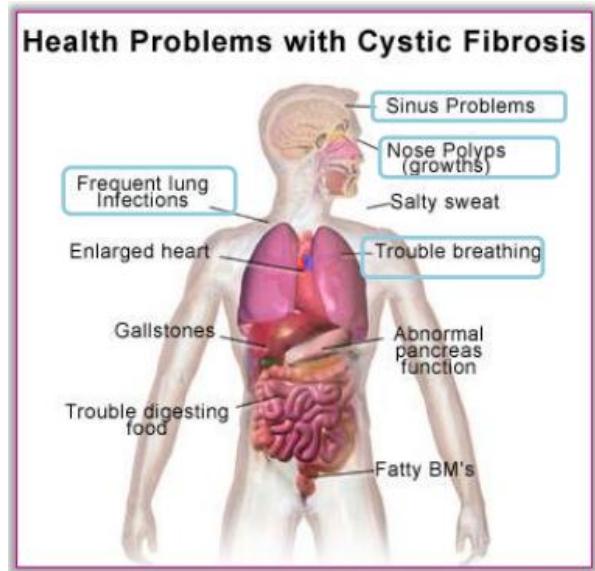
- **Males:** Congenital absence of vas deferens → infertility.
- **Females:** Thick cervical mucus → reduced fertility.

- **5. Sweat Glands**

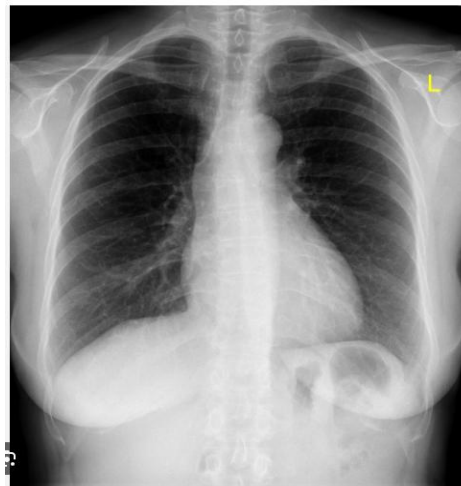
- CFTR dysfunction prevents reabsorption of **Na⁺ and Cl⁻** from sweat.
- Leads to:
 - **Salty sweat**
 - Positive **sweat chloride test** (≥ 60 mmol/L)



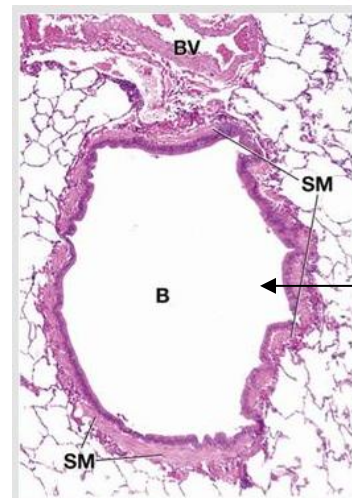
Cystic fibrosis, thoracic. Young man with a history of cystic fibrosis has hyperinflation and predominantly upper lobe bronchiectasis



<https://www.rareiseaseadvisor.com/hcp-resource/cystic-fibrosis-histology/>



Normal Chest X-Ray



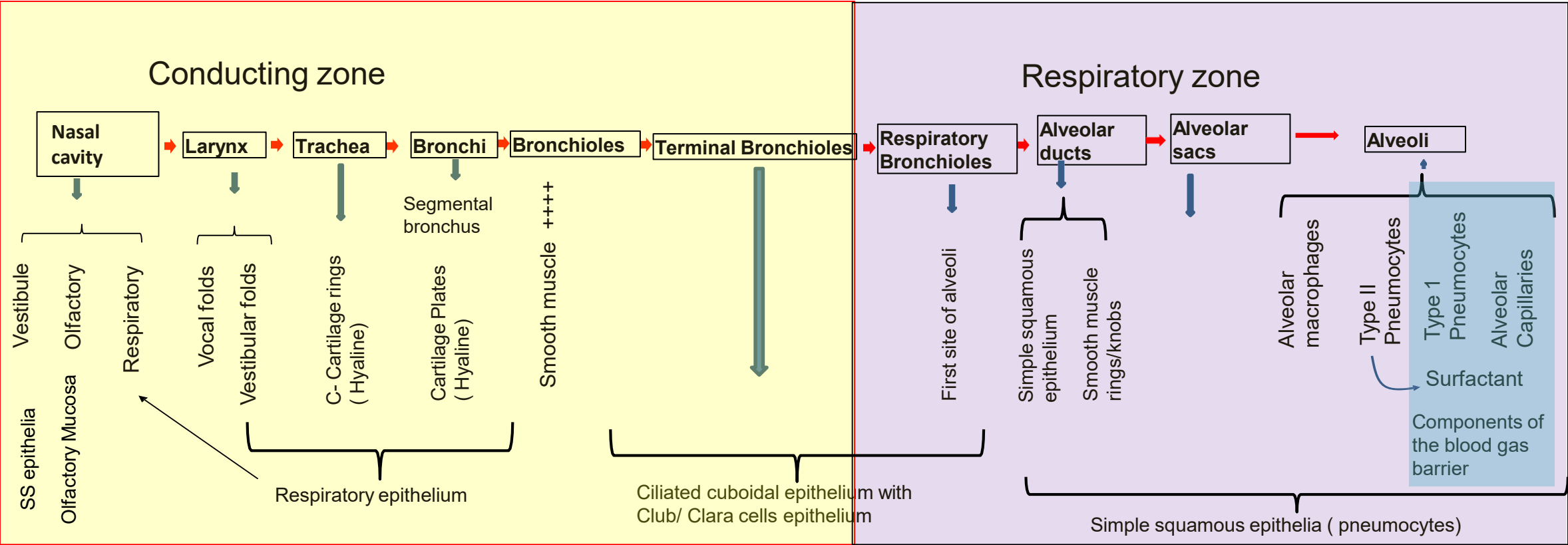
Normal

Ciliated Epithelium

Thickened mucus excretion

Bronchiole Lumen

Respiratory System - Summary of Structures



Support carried out by Cartilage → smooth muscle → elastic and reticular fibers

Decrease in goblet cells and cilia